



# Guaranteed Output Feedback Control for Uncertain Systems under Control and State Constraints

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**Abstract.** This paper deals with output feedback control of uncertain nonlinear dynamic systems in the presence of state and control constraints. We provide necessary and sufficient conditions for the guaranteed viability property defined by the existence of a Lipschitz closed-loop that maintains exactly the state of the system in a given closed domain of constraints despite some bounded disturbances acting both on the dynamics and on the output. Using the notion of Lipschitz kernel of a closed set-valued map, we obtain some equivalent geometric and Hamilton–Jacobi–Isaac conditions for the guaranteed viability property. We derive an algorithm to build a guaranteed viable output feedback.

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## 0. Introduction

Let us consider in  $\mathbb{R}^n$  the uncertain nonlinear dynamical system with imperfect information:

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t), w(t)), \\ y(t) &= h(x(t), w(t)), \\ u(t) &\in U(y(t)), \quad w(t) \in W(x(t)), \end{aligned} \quad (1)$$

where  $t \in \mathbb{R}_+$  is time,  $x(t) \in \mathbb{R}^n$  stands for the state of the system,  $y(t) \in \mathbb{R}^q$  stands for the output,  $u(t) \in \mathbb{R}^p$  is the control, and  $w(t) \in \mathbb{R}^m$  is a disturbance, representing the exogenous uncertainty. We assume here that the set  $U(y)$  of admissible controls may depend *a priori* on the observation  $y$  of the system and the set  $W(x)$  of possible disturbances may depend on the state  $x$  of the system.

We do not consider a statistical model on the exogenous uncertainty and, instead, adopt a pessimistic and total risk-averse approach [7] related to differential games. This worst case or ‘Min-Max’ or guaranteed framework has been used and developed for robust and  $H^\infty$  control by many authors mainly for stability or optimality issues. We refer to [9, 13] in the case of linear systems, to [18] for nonlinear systems and, in particular, to [5, 6, 23] for the use of viscosity solutions.

The aim of the present work is to study directly state constraints (or viability) problems

$$x(t) \in K, \quad \forall t \geq 0, \quad (2)$$

for controlled uncertain nonlinear systems of type (1) and especially to reveal single-valued output feedback law  $y \rightarrow u(y) \in U(y)$  such that, whatever the admissible perturbations  $w(\cdot)$  are, the solutions  $x(\cdot)$  of the system starting from  $K$  stay exactly in the set  $K$ . In other words, we aim at revealing, if it exists, a closed-loop control of the observation to ensure the invariance of the set  $K$  with respect to the disturbances. It appears that the invariance property is related to the Lipschitzianity of the dynamic and, consequently, of the feedback law. Consequently, as introduced in [2], we shall say that a set  $K$  is a guaranteed (or robust) viable domain for (1) if there exists an admissible Lipschitz feedback  $u(\cdot)$  such that the set  $K$  is invariant for the dynamics

$$\dot{x} = f(x, u(h(x, w))), \quad w \in W(x).$$

The present paper provides

- Necessary and sufficient conditions for this guaranteed viability property,
- Equivalent characterizations in a differential game framework,
- An algorithm to build guaranteed viable output feedbacks.

Compared to referenced works addressing the viability (or state constraints) problem, these results are new simultaneously for *feedback control*, *robustness* and *information* issues.

Indeed, most of the results obtained in the viability context [1] are related open-loop controls even in the deterministic case (i.e. without perturbation  $W(x) = \{w\}$ ) with perfect information ( $y = x$ ). We recall that a set  $K$  is said to be a viability domain [1] for a given control system  $\dot{x} = f(x, u)$ ,  $u \in U(x)$ , if and only if, for any  $x(0) \in K$ , there exists an absolutely continuous function  $x(\cdot)$  satisfying the viability condition (2) and a measurable control  $t \rightarrow u(t) \in U(x(t))$  such that

$$\dot{x}(t) = f(x(t), u(t)) \quad \text{for almost every } t \geq 0.$$

The problem of viable state feedback turns out to be mainly related to selection processes (single-valued function) of the set-valued ‘regulation map’

$$R_K(x) = \{u \in U(x), f(x, u) \in T_K(x)\},$$

where  $T_K(x)$  is the contingent cone at  $K$  in  $x$  defined by

$$T_K(x) = \left\{ v \in \mathbb{R}^n, D_{\uparrow} \mathbb{d}_K(x)(v) = \liminf_{h \rightarrow 0^+} \frac{\mathbb{d}_K(x + hv)}{h} \leq 0 \right\}.$$

Unfortunately, this set-valued map  $R_K$  is generally not regular enough to admit a relevant selection and consequently to reveal viable (single-valued) state feedback control; The situation is similar in optimal control issues whenever the value function is not regular enough to provide directly optimal synthesis through the Hamilton–Jacobi–Bellman equation [5, 8, 10]. Nevertheless, we mention that whenever this regulation map  $R_K$  is lower semi-continuous, the Michael Theorem [3] provides the existence of a continuous ‘viable’ feedback. Under similar assumptions, the so-called minimal selection (or slow solutions) [3, 16] of the regulation map gives also viable state feedback. We also mention that, whenever the dynamic  $f$  is Lipschitz, a geometrical condition involving the interior of the Clarke (or circa-tangent [4] or regular [20]) cone yields the existence of a Lipschitz closed-loop  $u(x)$  providing the viability of  $K$  (see for instance [3]). But this condition is only sufficient.

As far as imperfect information and viability problems are concerned, one can find some sufficient conditions in [24] but the described regulations have also a set-valued feature and the problem of selection arises similarly.

To deal with robustness issue and to extend state constraints results to the control of uncertain systems, the leadership and discriminating notions [2, 11] have been proposed and developed in the differential game framework. In particular, a closed set  $K$  is a discriminating ( $u/w$ ) domain for a system  $\dot{x} = f(x, u, w)$ ,  $u \in U$ ,  $w \in W$  if, for any  $x(0) \in K$ , there exists a nonanticipative strategy (see, for instance, [11])  $S_u: w(\cdot) \rightarrow u(\cdot)$ , such that, for any measurable  $w(\cdot)$ , the solution of

$$\dot{x}(t) = f(x(t), S_u(w)(t), w(t))$$

is viable in  $K$ , i.e. satisfies (2). In [11], P. Cardaliaguet provides equivalent geometric (or Hamiltonian) formulation for discriminating domains. He also introduced and characterized the discriminating kernel of a closed set  $K$  denoted by  $\text{Disc}_{(f,U,W)}^{(u/w)}(K)$  and defined by the largest closed discriminating domain contained in  $K$ . In [12], algorithms inspired from the viability kernel one are proposed to approximate this set. But in terms of robustness, this concept is not generally well-suited since the controller needs the knowledge of the perturbation  $w(t)$  at each instant. A more relevant tool for robustness is the leadership property. Namely, a set  $K$  is a leadership ( $u/w$ ) domain for a system  $\dot{x} = f(x, u, w)$ ,  $u \in U$ ,  $w \in W$  if, for any  $\varepsilon > 0$ , any  $T > 0$ , any  $x(0) \in K$ , and for any nonanticipative strategy  $S_w: u(\cdot) \rightarrow w(\cdot)$ , there exists a control  $u(\cdot)$  such that the solution of  $\dot{x}(t) = f(x(t), u(t), S_w(u)(t))$  is ‘quasi-viable’ in  $K$  in the sense that  $\text{d}_K(x(t)) \leq \varepsilon$  for every  $t \in [0, T]$ . The presence of this  $\varepsilon$  in the previous characterization is troublesome in the sense that we cannot guarantee exactly the viability although we can make the distance to  $K$  as small as possible (see for instance applications to stabilization under bounded chattering in [15]). As far as feedbacks are concerned, we can emphasize that the leadership property is associated with a piecewise constant state feedback of the form  $u(t, x)$  which is closely related to Krasowski–Subbotin

strategies [19]. But, to our knowledge, no general viability result is available in this differential game context with imperfect and/or partial information.

The present paper is organized as follows. In a first section, we precisely define the guaranteed viability property. Then, using the notion of Lipschitz kernel of a closed set-valued map [14] and an invariance characterization, we provide necessary and sufficient conditions for this guaranteed viability property. The Steiner selection allows to specify a relevant output feedback. In a second section, equivalent formulations are given in a differential game framework using discriminating characterizations. In section 3, we derive from the discriminating property an algorithm to compute relevant output feedback controls. Eventually, a simple numerical example attempts to illustrate the results.

## 1. Guaranteed Viability

Our first aim is to find necessary and sufficient conditions for the existence of a Lipschitz feedback law  $y \rightarrow u(y)$  that ensures the invariance of a given set  $K$  with respect to the disturbance  $w$  for the observed system (1). Such a property is referred as a guaranteed viability property (see [2] for a first definition in the perfect information case, i.e.  $y = x$ ).

To achieve the characterization of guaranteed viability, we use an invariance property and the Lipschitz kernel of a closed set-valued map.

### 1.1. DEFINITION OF GUARANTEED VIABILITY DOMAIN

We need to recall that a set  $K$  is said to be *invariant* under the differential inclusion  $\dot{x} \in \{g(x, w), w \in W(x)\}$ , if, for every  $x_0 \in K$ , every absolutely continuous solutions  $x(\cdot)$  of  $\dot{x} \in g(x, W(x))$ , starting from  $x_0$ , remains in  $K$ .

We also recall that a function  $u: \mathbb{R}^q \rightarrow \mathbb{R}^p$  is said to be a *selection* of a set-valued map  $U(\cdot): \mathbb{R}^q \rightsquigarrow \mathbb{R}^p$  if, for every  $y \in \mathbb{R}^q$ , we have  $u(y) \in U(y)$ . By a Lipschitz selection of  $U(\cdot)$ , we mean a selection  $u(\cdot)$  of  $U(\cdot)$  satisfying

$$|u(y_1) - u(y_2)| \leq k_u |y_1 - y_2|, \quad \forall y_1, y_2 \in \mathbb{R}^q.$$

Now we are ready to extend the definition of guaranteed viability domains to the case of imperfect and partial information as follows:

**DEFINITION 1.1.** Consider a set  $K \subset \mathbb{R}^n$ . We shall say that the set  $K$  is a guaranteed viability domain for the system (1) if and only if there exists a Lipschitz selection  $u(\cdot)$  of  $U(\cdot)$  such that  $K$  is invariant for the differential inclusion  $\dot{x} \in \{f(x, u(h(x, w)), w), w \in W(x)\}$ .

*Remark.* Whenever there is no perturbation, i.e. when  $W(\cdot)$  is reduced to a singleton, we obtain the definition of a ‘Lipschitz viability’ property in the sense that we require the existence of a Lipschitz feedback that ensures the viability of the constraint set  $K$ .

## 1.2. ASSUMPTIONS

A necessary and sufficient condition for a closed set  $K$  to be invariant under  $G$  lies in the following (epi-contingent) Hamilton–Jacobi condition [1]:

$$\forall x \in K, \quad \sup_{w \in W(x)} D_{\uparrow} \mathbb{d}_K(x)(g(x, w)) \leq 0,$$

or equivalently the geometric condition

$$\forall x \in K, \quad g(x, W(x)) \subset T_K(x),$$

whenever the dynamic  $G(x) = g(x, W(x))$  is Lipschitz on  $\mathbb{R}^n$  namely

$$\forall (x_1, x_2) \in \mathbb{R}^{n^2}, \quad G(x_1) \subset G(x_2) + \lambda |x_1 - x_2| \mathcal{B}_n,$$

with closed values. By  $\mathcal{B}_l$ , we mean the unit ball of  $\mathbb{R}^l$ .

Consequently, in order to use these invariance characterizations, we need to consider the following assumptions on the system (1) denoted by **Af**, **AU**, **AW** and **Ah**:

- *Assumptions **Af** on  $f$ :*

- (i) The function  $(x, u, w) \mapsto f(x, u, w)$  is Lipschitz on  $\mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p$ ,
- (ii) The function  $f_{x,w}: u \mapsto f(x, u, w)$  is convex for any  $(x, w)$  in the sense that

$$\forall R \subset \mathbb{R}^p, \forall x \in \mathbb{R}^n, w \in \mathbb{R}^m, \quad f(x, \text{co}(R), w) = \text{co}(f(x, R, w)),$$

where  $\text{co}$  represents the convex hull.

- *Assumption **AU** on  $U$ :* The set-valued map  $U: \mathbb{R}^q \rightsquigarrow \mathbb{R}^p$  has a closed graph and nonempty convex compact values.
- *Assumption **AW** on  $W$ :* The set-valued map  $W: \mathbb{R}^n \rightsquigarrow \mathbb{R}^m$  is Lipschitz on  $\mathbb{R}^n$  with nonempty compact values.
- *Assumption **AWbis** on  $W$ :* The set-valued map  $W: \mathbb{R}^n \rightsquigarrow \mathbb{R}^m$  is constant and the set  $W = W(x)$  is compact.
- *Assumption **Ah** on  $h$ :* The function  $h: \mathbb{R}^n \times \mathbb{R}^m \rightsquigarrow \mathbb{R}^q$  is Lipschitz on  $\mathbb{R}^n \times \mathbb{R}^m$ .

*Remark.* In particular, these assumptions hold true for affine systems of type

$$\begin{aligned} \dot{x} &= a(x, w) + b(x, w)u, \quad u \in U, w \in W, \\ y &= h(x, w), \end{aligned}$$

where the sets  $U$  and  $W$  are compact,  $U$  is convex and the functions  $a(\cdot, \cdot)$ ,  $b(\cdot, \cdot)$ ,  $h(\cdot, \cdot)$  are Lipschitz.

## 1.3. LIPSCHITZ KERNEL

As we mentioned, guaranteed viability characterizations require the regulation map to admit a Lipschitz selection. The notion of Lipschitz kernel of a set-valued map allows to deal with such an issue in a relevant way. We recall (see [14] in the same issue) that the  $\lambda$ -Lipschitz kernel of a set-valued map  $F: \mathbb{R}^q \rightsquigarrow \mathbb{R}^p$  with closed graph denoted by  $\mathcal{L}_\lambda(F)$  is the the largest (in the graphical sense)  $\lambda$ -Lipschitz set-valued map with closed graph contained in  $F$ .

In [14], a differential game (discriminating) characterization of the Lipschitz kernel is given and an algorithm inspired from the discriminating kernel one is also deduced to approximate this Lipschitz kernel. Indeed, the graph of the Lipschitz kernel is the discriminating kernel [11] of a suitable differential game namely

$$\text{Graph}(\mathcal{L}_\lambda(F)) = \text{Disc}_{(S^\lambda)}^{b/d}(\text{Graph}(F)),$$

where  $(S^\lambda)$  refers to the dynamics

$$\dot{y} = d, \quad d \in \mathcal{B}_q, \quad \dot{u} = \lambda b, \quad b \in \mathcal{B}_p. \quad (3)$$

## 1.4. CHARACTERIZATIONS OF GUARANTEED VIABILITY DOMAINS

Using invariance characterizations and the Lipschitz kernel of a specific set-valued map, we obtain the following necessary and sufficient characterization of closed guaranteed viability domains. We need to consider the set-valued map

$$R_K^\lambda(y) = \left\{ u \in U(y) \mid \forall x \in K, \forall w \in W(x), \exists b \in \mathcal{B}_p, \right. \\ \left. f(x, u + \lambda|y - h(x, w)|b, w) \in T_K(x) \right\},$$

or equivalently

$$R_K^\lambda(y) = \left\{ u \in U(y), \sup_{x \in K} \sup_{w \in W(x)} \inf_{b \in \mathcal{B}_p} D_{\uparrow \mathbb{D}_K}(x) \times \right. \\ \left. \times (f(x, u + \lambda|y - h(x, w)|b, w)) \leq 0 \right\}.$$

**THEOREM 1.2.** *We posit the assumptions  $Af$ ,  $AU$ ,  $AW$  and  $Ah$  defined in (1.2) on the system  $(f, U, W, h)$ . We consider a closed set  $K \subset \mathbb{R}^n$ . The following assertions are equivalent:*

- (i) *The set  $K$  is a guaranteed viability domain for the system (1),*
- (ii) *There exists  $\lambda \geq 0$  such that  $\forall y \in \mathbb{R}^q$ ,  $\mathcal{L}_\lambda(R_K^\lambda)(y) \neq \emptyset$ .*

*Remark 1.* Obviously, in the case of certain system ( $W$  is a singleton), we derive a more simple condition for the existence of a viable Lipschitz feedback.

*Remark 2.* The regulation maps  $R_K^\lambda(\cdot)$  and  $\mathcal{L}_\lambda(R_K^\lambda)(\cdot)$  reveal controls within the domain of constraints  $K$  and thus anticipate viability problems occurring on the boundary. This is a major difference compared to the leadership notion.

*Remark 3.* A necessary condition for guaranteed viability is clearly that, for every  $y \in \mathbb{R}^n$ ,  $R_K^\lambda(y) \neq \emptyset$ . For some examples, this regulation map  $y \rightsquigarrow R_K^\lambda(y)$  turns out to be Lipschitz which deeply simplifies the use of Theorem 1.2. At this stage, we do not know under which general assumptions this applies.

*Proof of Theorem 1.2.* First, we check that the set-valued map  $\mathcal{L}_\lambda(R_K^\lambda)(\cdot)$  is well-defined. We need the set-valued map  $R_K^\lambda(\cdot)$  to have a closed graph. Take any  $(y_n, u_n) \in \text{Graph}(R_K^\lambda)$  such that  $(y_n, u_n) \rightarrow (y, u)$ . Since  $u_n \in U(y_n)$  and  $U(\cdot)$  is closed, we obtain  $u \in U(y)$ . Moreover,

$$\forall x \in K, \forall w \in W(x), \exists b_n \in \mathcal{B}_p \text{ s.t.}$$

$$f(x, u_n + \lambda|h(x, w) - y_n|b_n, w) \in T_K(x).$$

Since the unit ball  $\mathcal{B}_p$  is compact, there exists a subsequence  $b_n$  converging into  $b \in \mathcal{B}_p$ . Since the tangent cone  $T_K(x)$  is closed and  $f$  is continuous, we deduce that

$$f(x, u + \lambda|h(x, w) - y|b, w) \in T_K(x).$$

This means that  $u \in R_K^\lambda(y)$  and  $\text{Graph}(R_K^\lambda)$  is closed.

– Now, let us assume (i) namely that  $K$  is a guaranteed viability domain for (1). Let us denote by  $u_\lambda(\cdot)$  the  $\lambda$ -Lipschitz selection of  $U(\cdot)$  involved in Definition 1.1. Then, from hypothesis **Af**, **AU**, **AW** and **Ah** on  $(f, U, W, h)$ , the dynamics  $f(x, u_\lambda(h(x, w)), w)$  satisfies the assumptions of the invariance theorem (see [1]). Consequently, the invariance of the set  $K$  under the differential inclusion

$$\dot{x} \in \{f(x, u_\lambda(h(x, w))), w, w \in W(x)\},$$

yields the geometrical conditions

$$\forall x \in K, \forall w \in W(x), \quad f(x, u_\lambda(h(x, w)), w) \in T_K(x).$$

Using the Lipschitzian feature of the feedback  $u_\lambda$ , we can claim, for every  $y \in \mathbb{R}^q$ , every  $x \in K$  and every  $w \in W(x)$ , the existence of  $b \in \mathcal{B}_p$  such that

$$u_\lambda(y) = u_\lambda(h(x, w)) + \lambda|h(x, w) - y|b.$$

Consequently, for every  $y \in \mathbb{R}^q$ , we have  $u_\lambda(y) \in R_K^\lambda(y)$ .

Furthermore, since  $u_\lambda(\cdot)$  is  $\lambda$ -Lipschitz on  $\mathbb{R}^q$ , using Proposition 3.2 in [14], we can claim that

$$\text{Graph}(u_\lambda) = \text{Graph}(\mathcal{L}_\lambda(u_\lambda)) \subset \text{Graph}(\mathcal{L}_\lambda(R_K^\lambda)).$$

In other words, for every  $y \in \mathbb{R}^q$ , we have  $u_\lambda(y) \in \mathcal{L}_\lambda(R_K^\lambda)(y)$  and we obtain (ii).

– Conversely, let us assume that the set-valued map  $R(\cdot) = \mathcal{L}_\lambda(R_K^\lambda)(\cdot)$  has nonempty values. Since the set-valued map  $R \subset U$  is  $\lambda$ -Lipschitz and has bounded values on  $\mathbb{R}^q$ , we can consider its Steiner (or curvature centroid) selection

$$u(y) = p \int_{\Gamma_p} v \sigma(R(y), v) \mu(dv),$$

where  $\sigma(R(y), v) = \sup_{u \in R(y)} \langle u, v \rangle$  is the support function of the set  $R(y)$ ,  $\Gamma_p$  stands for the unit sphere of  $\mathbb{R}^p$  and  $\mu$  is the measure on  $\Gamma_p$  proportional to the Lebesgue measure and satisfying  $\mu(\Gamma_p) = 1$ . Hence (see [4]),

- The feedback control  $u(\cdot): \mathbb{R}^q \rightarrow \mathbb{R}^p$  is  $p\lambda$ -Lipschitz on  $\mathbb{R}^q$ .
- For every  $y \in \mathbb{R}^n$ ,  $u(y) \in \text{co}(R(y)) \subset \text{co}(R_K^\lambda(y))$ . Since  $U(y)$  is convex, we deduce that  $u(y) \in U(y)$ . Consequently,  $u(\cdot)$  is a Lipschitz selection of  $U(\cdot)$ .
- Now take any  $x \in K$  and any  $w \in W(x)$ . By convexity assumption **AU**, we have

$$f(x, \text{co}(R_K^\lambda(h(x, w))), w) = \text{co}(f(x, R_K^\lambda(h(x, w))), w).$$

On the other hand, from the definition of  $R_K^\lambda(h(x, w))$ , we deduce that

$$f(x, R_K^\lambda(h(x, w)), w) \subset T_K(x).$$

Therefore, we obtain

$$\forall x \in K, \forall w \in W(x), \quad f(x, u(h(x, w)), w) \in \text{co}(T_K(x)).$$

From [4] (Theorem 4.1.10, p. 130), we have

$$\liminf_{z \xrightarrow{K} x} \text{co}(T_K(z)) \subset T_K(x).$$

Since, by assumptions Af, AW and Ah, the set-valued map  $F(x) = \{f(x, u(h(x, w)), w), w \in W(x)\}$  is Lipschitz, we deduce that

$$\forall x \in K, \quad F(x) \subset T_K(x).$$

Thus, using again the invariance theorem, we conclude that  $K$  is invariant for the dynamics  $\dot{x} \in F(x) = \{f(x, u(h(x, w)), w), w \in W(x)\}$ .

Therefore,  $K$  is a guaranteed viability domain for system (1).  $\square$

### 1.5. ROBUST VIABLE OUTPUT FEEDBACK

The proof of the previous theorem indicates how to build a robust feedback associated with a guaranteed domain  $K$  using the Steiner selection the regulation map  $\mathcal{L}_\lambda(R_K^\lambda)(\cdot)$ . Thus we define the output feedback:

$$u(y) = \text{Steiner}(R(y)) = p \int_{\Gamma_p} v \sigma(R(y), v) \mu(dv), \quad (4)$$

where

- $R(y) = \mathcal{L}_\lambda(R_K^\lambda)(y)$ ,
- $R_K^\lambda$  is defined in Theorem 1.2,
- $\sigma(R(y), v) = \sup_{u \in R(y)} \langle u, v \rangle$  is the support function of the set  $R(y)$ ,

- $\Gamma_p$  stands for the unit sphere of  $\mathbb{R}^p$  and  $\mu$  is the measure on  $\Gamma_p$  proportional to the Lebesgue measure and satisfying  $\mu(\Gamma_p) = 1$ .

**COROLLARY 1.3.** *We posit the assumptions **Af**, **AU**, **AW** and **Ah** defined in (1.2). If  $K$  is guaranteed for the system (1), then a Lipschitz robust viable output feedback is given by (4).*

## 2. Differential Game Characterizations

Whenever the set of admissible disturbances  $W(\cdot)$  does not depend on the state  $x$  (assumption **HWbis**), we can characterize equivalently the guaranteed property through a discriminating property [2, 11] of an extended system and in particular through a Hamilton–Jacobi–Isaacs relation. This result is deeply inspired from the discriminating characterization of the Lipschitz kernel (see again [14] in the same issue) and plays a key role in the algorithm defined in the next section of the paper. We need to recall that  $\text{NP}_C(x)$  denotes the set of proximal normals to the set  $C$  at  $z \in C$  defined by

$$p \in \text{NP}_C(z) \Leftrightarrow d_C(z + p) = \|p\|.$$

**THEOREM 2.1.** *Assume hypothesis **Hf**, **HU**, **HWbis** and **Hh** defined in 1.2 on the dynamics  $(f, U, W, h)$ . Consider a closed set  $K \subset \mathbb{R}^n$ . Consider the set-valued map  $R_\lambda(\cdot) = \mathcal{L}_\lambda(R_K^\lambda)(\cdot)$  with  $R_K^\lambda$  defined as in Theorem 1.2.*

*The following assertions are equivalent:*

- (i) *The set  $K$  is guaranteed for the system  $(f, U, W, h)$ ,*
- (ii) *There exists  $\lambda \geq 0$  such that the set  $K \times \text{Graph}(R_\lambda)$  is a discriminating  $(b_1, b_2)/(d, w)$  domain for the system*

$$\begin{cases} \dot{x} = f(x, u + \lambda|h(x, w) - y|b_1, w), & w \in W, b_1 \in \mathcal{B}_p, \\ \dot{y} = d, & d \in \mathcal{B}_q, \\ \dot{u} = \lambda b_2, & b_2 \in \mathcal{B}_p. \end{cases} \quad (5)$$

- (iii) *There exists  $\lambda \geq 0$  such that*

$$\begin{aligned} & \forall x \in K, \forall (y, u) \in \text{Graph}(R_\lambda), \\ & \forall w \in W, \forall d \in \mathcal{B}_q, \exists (b_1, b_2) \in \mathcal{B}_p^2, \\ & f(x, u + \lambda|h(x, w) - y|b_1, w) \in T_K(x), \\ & (d, \lambda b_2) \in T_{\text{Graph}(R_\lambda)}(y, u). \end{aligned} \quad (6)$$

- (iv) *There exists  $\lambda \geq 0$  such that*

$$\begin{aligned} & \forall x \in K, \forall (y, u) \in \text{Graph}(R_\lambda), \\ & \forall p \in \text{NP}_K(x), \forall (q, r) \in \text{NP}_{\text{Graph}(R_\lambda)}(y, u), \\ & H(p, q, r, x, y, u) \leq 0, \end{aligned} \quad (7)$$

where the Hamiltonian  $H$  is defined by

$$H(p, q, r, x, y, u) = \sup_{w \in W} \inf_{b \in \mathcal{B}_p} \langle p, f(x, u + \lambda|h(x, w) - y|b, w) \rangle + |q| - \lambda|r|,$$

*Proof.*

- The equivalence between (ii), (iii) and (iv) is a direct consequence of the discriminating property (see again [11]). In particular, the Hamiltonian of (iv) is obtained through the relation

$$\begin{aligned} & \sup_{w \in W, d \in \mathcal{B}_q} \inf_{b_1 \in \mathcal{B}_p, b_2 \in \mathcal{B}_p} \langle (p, q, r), (f(x, u + \lambda|h(x, w) - y|b_1, w), d, \lambda b_2) \rangle \\ &= \sup_{w \in W} \inf_{b \in \mathcal{B}_p} \langle p, f(x, u + \lambda|h(x, w) - y|b, w) \rangle + |q| - \lambda|r|. \end{aligned}$$

Thus it is enough to prove that (ii)  $\Rightarrow$  (i)  $\Rightarrow$  (iii).

- Indeed, assume that (ii) holds. Clearly, we derive that

$$\forall y, R_\lambda(y) = \mathcal{L}_\lambda(R_K^\lambda)(y) \neq \emptyset.$$

By Theorem 1.2, we obtain that  $K$  is guaranteed namely (i).

- Conversely, assume that (i) holds true. By Theorem 1.2,  $R_\lambda$  has nonempty values. Now, take any  $x \in K$ , any  $(y, u) \in \text{Graph}(R_\lambda)$ , any  $w \in W$  and any  $d \in \mathcal{B}_q$ .

Since  $R_\lambda$  is  $\lambda$ -Lipschitz, Proposition 2.1 in [14] first yields

$$\exists b_2 \in \mathcal{B}_p, \quad (d, \lambda b_2) \in T_{\text{Graph}(R_\lambda)}(y, u). \quad (8)$$

Moreover, since  $R_\lambda(y) = \mathcal{L}_\lambda(R_K^\lambda)(y) \subset R_K^\lambda(y)$ , from the very definition of  $R_K^\lambda$ , we obtain the existence of  $b_1 \in \mathcal{B}_q$  s.t.

$$f(x, u + \lambda|h(x, w) - y|b_1, w) \in T_K(x)$$

and thus (iii) holds true. This concludes the proof.  $\square$

### 3. An Algorithm for the Guaranteed Lipschitz Feedbacks

We use Theorem 2.1 and adapt the discriminating (and viability) kernel algorithm to compute the set-valued map  $\mathcal{L}_\lambda(R_K^\lambda)$  of Lipschitz feedbacks as follows. We consider the set sequence below:

$$\begin{aligned} R_0(y) &= U(y), \\ R_{k+1}(y) &= \{u \in R_k(y) \text{ s.t. } \forall x \in K, \\ &\quad (x, y, u) \in \text{Disc}_{(S)}(K \times \text{Graph}(R_k))\}, \end{aligned} \quad (9)$$

where  $(S)$  refers to the differential game (5) defined in Theorem 2.1.

We obtain the convergence of the maps  $R_k(\cdot)$  to the Lipschitz regulation map  $\mathcal{L}_\lambda(R_K^\lambda)(\cdot)$  in the following graphical sense.

**PROPOSITION 3.1.** *We assume hypothesis Hf, HU, HWbis and Hh defined in Theorem 1.2 on system  $(f, U, W, h)$ . Consider a closed set  $K \subset \mathbb{R}^n$ .*

*If  $K$  is guaranteed for system  $(f, U, W, h)$ , then there exists  $\lambda \geq 0$  such that*

$$\lim_{k \rightarrow \infty} \text{Graph}(R_k) = \bigcap_{k \in \mathbb{N}} \text{Graph}(R_k) = \text{Graph}(\mathcal{L}_\lambda(R_K^\lambda)),$$

where the sequence of set-valued map  $R_k(\cdot)$  is defined by (9) and (5).

*Remark.* We have to point out that this algorithm requires high dimensions in the sense we have to solve a differential game problem in dimension  $n + q + p$ . Even in the deterministic case with perfect information, this appears the price to pay to solve in full generality such a problem of viable Lipschitz feedback.

*Proof.* Let us set

$$\text{Graph}(R_\infty) = \bigcap_{k \in \mathbb{N}} \text{Graph}(R_k),$$

and prove that  $\text{Graph}(R_\infty) = \text{Graph}(\mathcal{L}_\lambda(R_K^\lambda))$ .

- (a) We know that the set-valued map  $U = R_0$  has a closed graph. Since the discriminating kernel of a closed set is closed, it turns out in a recursive way that the sets  $\text{Graph}(R_k)$  are well-defined and closed. Consequently, the set  $\text{Graph}(R_\infty)$  is closed.
- (b) Now, let us prove in a recursive way that  $\mathcal{L}_\lambda(R_K^\lambda) \subset R_\infty$ .  
We have clearly  $\mathcal{L}_\lambda(R_K^\lambda) \subset R_0 = U$ . Now assume for some  $k \in \mathbb{N}$  that  $\mathcal{L}_\lambda(R_K^\lambda) \subset R_k$ . Since  $K$  enjoys the guaranteed viability property, Theorem 2.1 yields

$$\begin{aligned} K \times \text{Graph}(\mathcal{L}_\lambda(R_K^\lambda)) &= \text{Disc}_{(S)}(K \times \text{Graph}(\mathcal{L}_\lambda(R_K^\lambda))) \\ &\subset \text{Disc}_{(S)}(K \times \text{Graph}(R_k)), \end{aligned}$$

which means that  $\mathcal{L}_\lambda(R_K^\lambda) \subset R_{k+1}$ . Therefore  $\mathcal{L}_\lambda(R_K^\lambda) \subset R_\infty$ .

- (c) Now, let us prove the converse inclusion. The discriminating property is ‘stable’ [11] in the sense that

$$\limsup_{k \rightarrow \infty} \text{Disc}(M_k) \subset \text{Disc}(\limsup_{k \rightarrow \infty} (M_k)).$$

Consequently, we obtain

$$K \times \text{Graph}(R_\infty) \subset \text{Disc}_{(S)}(K \times \text{Graph}(R_\infty)).$$

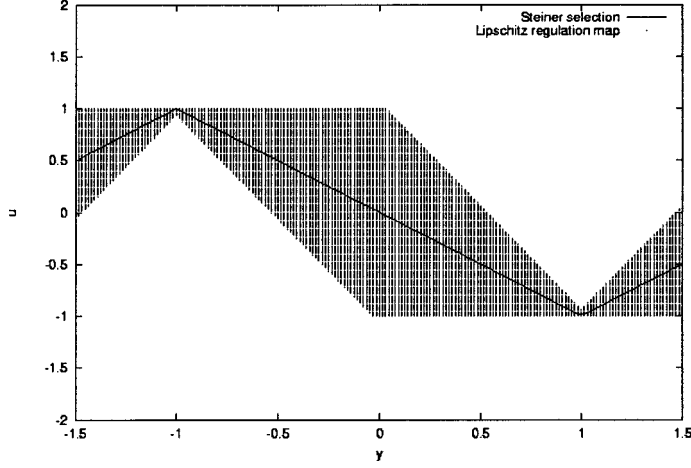
This means that  $K \times \text{Graph}(R_\infty)$  is a discriminating domain for the differential game (3) and we obtain

$$\forall x \in K, \forall (y, u) \in \text{Graph}(R_\infty), \forall w \in W, \forall d \in \mathcal{B}_q, \exists (b_1, b_2) \in \mathcal{B}_p^2, \\ (f(x, u + \lambda|h(x, w) - y|b_1, w), d, \lambda b_2) \in T_{K \times \text{Graph}(R_\infty)}(x, y, u).$$

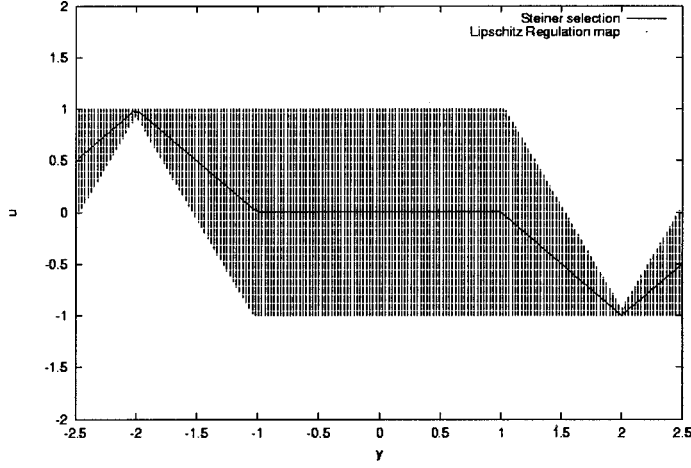
Since

$$T_{K \times \text{Graph}(R_\infty)}(x, y, u) = T_K(x) \times T_{\text{Graph}(R_\infty)}(y, u),$$

the very definition of  $R_K^\lambda$  yields  $R_\infty \subset R_K^\lambda$ . Moreover, from the Lipschitz characterization (2.1) in [14], we obtain that  $R_\infty$  is  $\lambda$ -Lipschitz. In other words, we have  $R_\infty \subset \mathcal{L}_\lambda(R_K^\lambda)$ . This concludes the proof.  $\square$



(a)



(b)

Figure 1. The graphs of both the Lipschitz regulation map  $\mathcal{L}_2(R_K^2)$  and its Steiner selection for the dynamics  $\dot{x} = u + w$ ,  $u \in [-1, 1]$ ,  $w \in [-1, 1]$  and the constraint  $K = [-1, 1]$ . (a) Perfect information:  $y = x$ . (b) Perfect information:  $y = x + w$ .

#### 4. A Numerical Example

We use the algorithm (9) to treat (in  $\mathbb{R}$ ) the academic example

$$\dot{x} = u + w, \quad u \in [-1, 1], \quad w \in [-1, 1], \quad (10)$$

and the constraint is  $K = [-1, 1]$ .

*Perfect information.* We consider the perfect observation of the system namely  $y = x$ . Using the algorithm (9), we approximate the graph of the Lipschitz regulation map  $\mathcal{L}_2(R_K^2)$ . Figure 1(a) shows that the Steiner selection is given by the projection on  $K$ , i.e.  $u(y) = -\Pi_{[-1,1]}(y)$ .

*Imperfect information.* Now, we consider the imperfect observation  $y = x + w$ . Figure 1(b) shows an approximation of  $\mathcal{L}_2(R_K^2)$  and the Steiner selection given by

$$u(y) = \begin{cases} 0, & \text{if } y \in [-1, 1], \\ 1 - |y + 2|, & \text{if } y \in [-3, -1], \\ -1 + |2 - y|, & \text{if } y \in [1, 3]. \end{cases}$$

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